

# Electric Vehicle Charging Demand Prediction using Machine Learning Algorithms

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## Abstract-

The increasing concerns regarding climate change, notably the surge in greenhouse gas emissions, have attained paramount significance in the agendas of both U.S. and global policymakers. Transportation, responsible for 29% of U.S. greenhouse gas emissions according to the U.S. Environmental Protection Agency (EPA), stands as the nation's leading contributor to this challenge. In parallel, substantial policy initiatives at the national and state levels are underway, and key automotive manufacturers like Ford, GM, and Volvo have committed to a complete transition to electric vehicles (EVs) within the next decade, heralding a profound shift in the automotive industry. This impending transformation, combined with public and private sector efforts, is poised to generate a substantial increase in EV demand. Governments and public utilities can use this research to build and manage public charging networks strategically by applying machine learning models to forecast energy consumption patterns inside EV charging infrastructure. By proactively forecasting future EV charger energy consumption, this study aims to pinpoint regions of high demand, facilitating optimized infrastructure expansion and resource utilization, aligning with sustainability goals and the evolving EV landscape. We harnessed a real-time dataset comprising 433,172 electric vehicle transactions, including 91,255 new vehicle registrations. We integrated data from the National Renewable Energy Laboratory's API, encompassing 1,686 charging stations in Washington. Our modeling process, applied uniformly across counties, included seasonal decomposition, training and testing data partitioning, gridsearch with pmdarima for parameter optimization, and SARIMAX modeling for forecasting. Visualization aided in assessing forecast accuracy, contributing to our overall study on EV charging demand prediction.

**Index Terms-** Electric Vehicle(s), Electric Vehicle Charging, ARIMA, Time-series Modelling, Machine Learning

## I. INTRODUCTION

Concerns about climate change and growing greenhouse gas emissions have become more widespread in recent years, and in many nations, the transportation industry is responsible for a sizable 29% of these emissions. Governments worldwide, following the example of the United States under the Biden administration, are actively prioritizing clean energy initiatives and the development of comprehensive electric vehicle (EV) charging networks to address this challenge [1]. Additionally, some states, like California and Washington, are taking independent actions to phase out gasoline-powered vehicles by 2035 and 2030, respectively. Simultaneously, major automotive manufacturers are committing to full electrification, signaling a profound transformation in the automotive industry [2]. These collective efforts are expected to drive a surge in global EV demand, raising critical questions about the readiness and capacity of existing charging infrastructure to meet this growing demand. The Washington Department of Transportation has set aside \$8 million in funding for projects related to charging infrastructure between July 1, 2021, and June 30, 2023, in order to solve this. Governments and EV charging providers can benefit greatly from this in terms of business. But predicting and modeling the demand for electric vehicles in various locations with accuracy is the main challenge.

This paper focuses on applying machine learning models to forecast the proliferation of electric vehicles with the highest potential for new acquisitions, offering strategic investment recommendations for optimal resource allocation. The scope of this project includes exploring predictive modeling techniques, analyzing data sources and methodology for EV demand prediction, understanding regional variations in EV adoption, and providing recommendations for efficient infrastructure placement to meet growing EV demand while promoting sustainable electric mobility solutions. In summary, this project aims to utilize data from Washington State's API and the National Renewable Energy Laboratory's API to forecast EV demand, offering valuable insights for charging infrastructure allocation amidst increasing EV adoption.

## II. LITERATURE SURVEY

While there has been progress in the field of predicting the demand for electric vehicle (EV) charging, there are still certain important factors that are generally disregarded, like real-world traffic distribution data and meteorological conditions in prediction models [1]. In order to close this gap, this study offers a thorough technique that incorporates weather insights and rich historical traffic data from South Korea into the forecasting model.

The proposed methodology unfolds in multiple phases to ensure robustness and accuracy. Initially, a sophisticated cluster analysis is employed to categorize intricate traffic patterns, followed by a meticulous relational analysis to pinpoint influential factors affecting charging demand. Then, a decision tree structure is used to provide exact classification criteria, paying special attention to variables such as the charging start time, which is calculated based on real traffic patterns, and the initial state of charge of EV batteries. These intricately woven components form the backbone of the model, enabling it to capture nuanced variations in charging load profiles across different temporal and spatial contexts, such as weekdays versus weekends and seasonal variations in summer and winter, while also delineating the disparities between residential and commercial charging sites.

Using real-world datasets from several locales, including the State of Utah and the City of Los Angeles, the suggested model is thoroughly validated [2]. The one-step-ahead prediction and the more difficult multi-step-ahead prediction are the two main prediction targets covered by this validation. The model's effectiveness is demonstrated by rigorous benchmarking and comparison against a variety of time series and machine learning models. Seq2Seq stands out as a particularly strong performer, excelling in multi-step predictions, indicating its proficiency with complex sequential data structures.

Furthermore, the research explores new paradigms for predicting energy consumption in EV networks by utilizing state-of-the-art machine learning techniques [3]. An accurate prediction of energy demand can be achieved by a charging station provider (CSP) assimilating data from numerous charging stations (CSs) through the use of the robust Energy Demand Learning (EDL) algorithm. A novel Federated Energy Demand Learning (FEDL) approach is presented to overcome possible issues about data privacy and communication overhead. This method significantly reduces communication overhead while protecting data privacy for EV users by allowing CSs to communicate just trained models with the CSP for processing, facilitating data sharing without compromising privacy.

Additionally, a time-spatial EV charging-power demand forecast model designed specifically for fast-charging stations located in cities is released [4]. This approach goes into the domain of predicting demand for urban EV charging, which sets it apart from earlier research that concentrated mostly on private charging stations with set charging-start timings. It uses cutting-edge methods like teleportation and Markov-chain traffic modelling to properly anticipate EV arrival rates and the ensuing

demand for charging power at public fast-charging stations. It does this by utilizing real-time closed-circuit video (CCTV) data from genuine metropolitan road networks. The model's ability to predict charging-power demand profiles over a range of time intervals is demonstrated numerically in the examples provided, which adds value to the planning of investments and operations for adaptive EV charging infrastructures in urban environments. This is particularly true with regard to energy storage optimization and renewable resource optimization based on dynamic patterns in EV charging demand.

Using information from a large dataset of more than 76,000 private EV trajectories in Beijing in January 2018, this study presents a novel Long Short-Term Memory (LSTM) neural network designed for forecasting electric vehicle (EV) charging demand at the station level over brief time intervals [5]. The Auto-Regressive Moving Average (ARIMA) and Multiple Layer Perceptron (MLP) models are two well-known time series prediction techniques that are used to systematically evaluate the performance of the LSTM model. By means of comprehensive scenario analyses that take into account variations in input data structures, sample sizes, time spans, and intervals, the research demonstrates the superiority of the LSTM model, as demonstrated by noticeably lower Mean Absolute Percentage Error (MAPE) values (6.83% for LSTM, 21.58% for ARIMA, and 18.31% for MLP). Notably, the research underscores the crucial role of time span and interval settings in optimizing the LSTM model's predictive accuracy, particularly favoring shorter intervals such as 1 hour.

This research fills in the gaps found in previous studies by using extensive historical traffic and meteorological data from South Korea to create a reliable forecasting model for the demand for EV charging [6]. The methodological framework integrates sophisticated techniques like cluster analysis for traffic pattern categorization, relational analysis for influential factor identification, and decision tree modelling for establishing classification criteria. Important factors like the initial battery state of charge and the charging initiation time are carefully taken into account. This is reinforced by case studies that provide illustrative examples of different charging load profiles in various temporal and spatial contexts, such as residential and commercial settings and weekdays and weekends. This suggested model serves as a basic tool for future study analysing the larger impact of EV charging on the power system infrastructure, in addition to helping power system engineers anticipate EV charging demand based on historical traffic and weather data.

In the realm of EV charging station energy consumption prediction, this study delves into the development of an efficient core cell phone application algorithm, specifically targeting UCLA's EV charging stations [7]. With a paramount focus on real-time predictions within seconds, the research evaluates and selects the k Nearest Neighbor (kNN) algorithm with k=1 for its superior performance among four fast machine learning-based time series prediction algorithms. The resulting cell phone application offers users predictive insights into expected available energy levels at the station and estimated charging completion times, achieving remarkable processing speeds of about one second. This streamlined prediction engine not only

enhances user experience but also contributes significantly to operational efficiency at EV charging stations, optimizing resource allocation and user convenience.

To further improve EV charging demand forecasting, this study presents a novel deep learning-based Long Short-Term Memory (LSTM) recurrent neural network model [8]. Using an arithmetic optimization algorithm (AOA) and data decomposition methods such as empirical mode decomposition (EMD), the new EMD-AOA-DLSTM neural predictor carefully tunes parameters to overcome the drawbacks of simple recurrent neural networks, particularly reducing problems with vanishing and exploding gradients. The model's performance is validated using real-world EV charging datasets from Georgia Tech, Atlanta, achieving remarkable prediction accuracy rates of 97.14% with minimal mean absolute and square errors. These findings affirm the efficacy of the deep learning LSTM neural predictor in delivering precise and reliable forecasts for EV charging demand, establishing a new benchmark in the field of predictive analytics for electric vehicle infrastructure planning and management.

An extensive analysis of real-world data-driven electric vehicle (EV) control charging schemes is provided [9]. The three main kinds of EV control charging systems that this paper explores are scheduling, clustering, and forecasting strategies. A detailed analysis of each strategy is provided, emphasizing the models and methods used for EV charging control. The objective of the comparative analysis is to assess the efficacy, benefits, and drawbacks of different tactics, with an emphasis on how well they work in actual situations. This review heavily relies on machine learning and probability-based techniques to provide insight into how EV charging management is developing.

Moreover, this study also addresses a number of other issues and difficulties that came up during the development of EV charging control, particularly in relation to next-generation smart grid applications. It offers insightful analysis and practical suggestions for resolving these issues and improving the efficacy and efficiency of EV charging techniques. Additionally, the paper offers a roadmap for future studies in this research domain, emphasizing the need for standardized guidelines and methodologies to ensure comparability and consistency across different research findings. These insights collectively contribute to the ongoing efforts aimed at advancing EV charging methods and demand-side management (DSM) in the context of smart grid applications, paving the way for a more sustainable and efficient electric vehicle ecosystem.

In the area of predictive analytics, a creative study suggests a deep learning model for EV charging demand prediction that is Transformer-based [10]. The impetus for this work comes from the shortcomings of recurrent neural networks (RNNs/LSTMs) and standard time series approaches, which may find it difficult to accurately capture intricate linkages and long-term dependencies. The Transformer model, known for its ability to handle sequential data with parallel processing, offers a promising solution to these challenges. The study aims to evaluate the Transformer model's performance using real-world data spanning short-term (7 days) and long-term (30, 60, 90 days) prediction horizons. Performance metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE)

will be utilized for a comprehensive comparison against existing methodologies. The anticipated outcomes of this research include insights into the Transformer model's efficacy in capturing intricate EV charging demand patterns and its potential to outperform traditional approaches.

In the realm of electric vehicle (EV) battery management, accurate prediction of the battery state of charge (SoC) is paramount [11]. Traditional methods often fall short in capturing the complex interplay between various factors influencing SoC, leading to suboptimal management decisions. This paper introduces a novel approach leveraging graph convolutional networks (GCNs) for EV battery SoC prediction. GCNs are renowned for their ability to capture intricate data relationships, making them well-suited for SoC prediction tasks. The model incorporates learnable adjacency matrices, which encode the relationships between battery/driving data points and temporal relationships between timestamps. Real-world EV data is utilized for training and validation, ensuring the model's accuracy and reliability in real-world scenarios. The GCN-based model's performance is evaluated against traditional recurrent neural network (RNN) baselines, demonstrating significant improvements in SoC prediction accuracy. Furthermore, the learnable adjacency matrices offer interpretability, allowing researchers and practitioners to gain insights into the most significant variables impacting SoC. This research contributes to advancing the state-of-the-art in EV battery management, with implications for optimizing charging strategies and enhancing overall EV performance and longevity.

Lastly, a study investigates the feasibility of predicting EV charging demand within a regulated electricity market context using deep learning techniques [12]. This research is based on actual data from Moroccan public charging stations, which covers a range of charging situations and consumption trends. There is an evaluation of four deep learning models: gated recurrent units (GRUs), long short-term memory (LSTM), recurrent neural networks (RNNs), and artificial neural networks (ANNs). The study employs performance metrics like mean absolute percentage error (MAPE) and root mean squared error (RMSE) to evaluate how well these models can predict outcomes. The findings reveal that all four models exhibit good accuracy in predicting EV charging demand, highlighting the potential of deep learning in this domain. However, the single hidden layer GRU model stands out, showcasing superior performance metrics with lower RMSE and MAPE values during both training and testing phases. These results underscore the suitability of GRUs for capturing time-series data and time-dependent patterns inherent in EV charging demand. The study also acknowledges the limitations posed by the relatively small dataset size and emphasizes the need for further research on smart charging strategies to optimize grid integration of EVs, particularly considering Morocco's significant renewable energy potential. Future work could explore charging behavior across diverse locations and delve into innovative approaches such as vehicle-to-grid (V2G) technologies for more efficient energy management during peak demand periods.

A study published in Energy Informatics delves into the evaluation of various machine learning (ML) and deep learning (DL) models for forecasting electric vehicle (EV) charging demand across different aggregation levels [13]. The researchers

conducted a comprehensive analysis, comparing the performance of ML models such as Ada Boosting and Random Forest with DL models. Their findings revealed that while ML models like Ada Boosting and Random Forest provided reliable forecasts, DL models showcased potential for superior accuracy. However, it was noted that DL models often required greater computational resources due to their complex architecture and training requirements. The significance of precise charging demand prediction cannot be overstated, particularly for grid operators and energy suppliers looking to seamlessly integrate EVs into the power grid and optimize energy procurement strategies. This research contributes significantly by providing a valuable framework for evaluating and selecting appropriate forecasting models, thereby advancing the objective of efficient EV charging demand prediction and management.

In a recent study addressing the pressing need for optimized electric vehicle (EV) charging infrastructure, researchers proposed a novel physics-informed and attention-based graph learning approach for regional EV charging demand prediction [14]. The proposed approach, named Physics-Informed Attention-based Graph (PAG), is designed to overcome the limitations of purely data-driven models by integrating multiple advanced techniques. PAG utilizes a Graph Attention Network (GAT) to capture spatial relationships between charging stations, emphasizing impactful connections crucial for accurate demand prediction. Additionally, a Temporal Pattern Attention (TPA) mechanism is incorporated to prioritize relevant historical data, enhancing forecasting accuracy. To ensure a nuanced understanding of data relationships and mitigate potential misinterpretations, PAG employs Physics-Informed Meta-Learning (PIML) during pre-training. PIML leverages knowledge of physical and economic laws to guide the model towards a more accurate representation of these relationships. The PAG model comprises three key modules: a Graph Embedding Module utilizing convolutional neural networks (CNNs) and multi-head GAT for feature extraction, a Multivariate Decoder Module incorporating TPA-LSTM for data processing, and a Model Pre-training Module employing meta-learning with physics and economic law-based tuning samples. This comprehensive approach positions PAG as a promising solution for accurate and interpretable regional EV charging demand prediction, thereby contributing significantly to the optimization of EV charging infrastructure planning and management strategies.

Taking a novel perspective on EV charging behavior prediction, researchers propose an innovative approach that integrates weather, traffic, and local event data alongside historical charging records [15]. This approach builds upon previous studies that primarily relied on past charging information alone. The research team collected a dataset comprising charging sessions from a university campus and enriched it with additional contextual data such as weather conditions, traffic patterns, and local events. Various machine learning algorithms including random forest, support vector machine (SVM), XGBoost, and deep neural networks were employed to predict session duration and energy consumption accurately. The results demonstrated that incorporating additional data sources led to significantly improved prediction accuracy compared to models relying solely on historical charging data. This study underscores the value of machine learning techniques in optimizing EV charging station efficiency by enhancing the prediction of charging behavior based on a comprehensive set of contextual factors.

### III. PROPOSED METHODOLOGY

The proposed methodology begins with the extraction of data from the Washington State's API, encompassing various parameters relevant to electric vehicle (EV) charging demand. The retrieved data undergoes a meticulous cleaning process to address any inconsistencies or missing values, ensuring the integrity and reliability of the dataset. Following this, an in-depth Exploratory Data Analysis (EDA) is conducted to gain valuable insights into the patterns and trends within the data, aiding in the formulation of predictive models. Seasonal decomposition, which enables to separate the time series data into its seasonal, trend, and residual components, is one of the primary modeling approaches used in our methodology. This decomposition process is crucial for understanding the underlying patterns and fluctuations in EV charging demand across different countries within Washington State.

An essential aspect of our methodology is the utilization of SARIMAX (Seasonal Autoregressive Integrated Moving Average with Exogenous Factors) models. SARIMAX models are advanced time series models that incorporate both autoregressive and moving average components, along with the ability to incorporate exogenous variables.

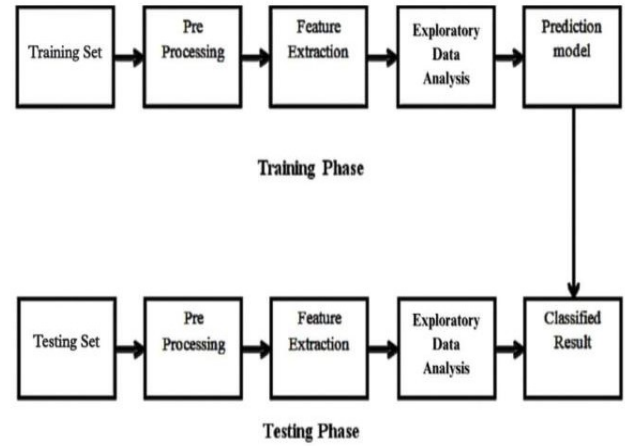


Fig. 1: Architecture Diagram

#### A. Dataset

The project utilized Washington State's API to collect data on electric vehicle title and registration activities, totalling 433,172 transactions (including 91,255 new vehicle registrations). Additionally, the National Renewable Energy Laboratory's API provided information on the existing 1,686 charging stations in Washington. This study accessed API keys and tokens, retrieved and parsed the data into a dictionary format, and finally saved it in a compressed .csv file for analysis

#### B. Data Cleaning

The acquired data performs a cleaning process, including feature engineering. The 'transaction\_date' column was converted from an object to a datetime format to serve as the dataset's index.



Columns pertaining to government-specific details, such as fees and tax credits, were deemed irrelevant to the project's objectives and were consequently removed. Redundant columns were also eliminated to streamline the dataset. Following these steps, the dataset, devoid of registration information, was scrutinized for duplicate entries, which were then carefully examined and removed. Furthermore, null values were identified and appropriately handled.

### C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted alongside data cleaning to guide feature selection and transformation. Our primary goal was to forecast the increase in new EV adoption for each county over time. Visualizations generated during the EDA process revealed a significant, decade-long surge in EVs within Washington State. This finding underscores the substantial demand for EVs in the state, presenting a lucrative opportunity for companies involved in EV charging infrastructure.

### D. Model Building

We divided the county-specific data into training and testing sets. To check for seasonality and examine trend information, the modelling technique will be the same for every county and will use seasonal decomposition. Various models including linear regression, random forest, bagging and boosting will be used and compared. Each model's performance will be evaluated.

### E. Model Selection and Evaluation

We opted for time series modeling to predict future EV adoption in each county. Various candidate models, including Seasonal ARIMA (SARIMA), linear regression, random forest, bagging, and boosting, were considered for evaluation. The performance of each model was assessed using a suite of statistical metrics.

### F. Data Splitting and Parameter Optimization

The data for each county was partitioned into training and testing sets. Prior to modeling, a seasonal decomposition analysis was conducted to discern any underlying seasonality and trend patterns within the data. This analysis confirmed the presence of a pronounced seasonal component.

The training and testing sets were established to facilitate model evaluation. An 80/20 split ratio (80% training, 20% testing) was identified as optimal through iterative testing for the initial county (King County).

The "pmdarima" library was employed to programmatically optimize the model parameters for each county. These optimized parameters were then incorporated into a SARIMA model, forming the basis for our predictive model. The model's performance was validated by comparing its generated forecasts with the actual data in the testing set.

### G. Model Validation and Future Predictions

The model was validated by generating forecasts for the testing data using a dedicated function. The results indicated that the

model possesses the capability to make accurate forecasts for future periods. The observed data points fell well within the confidence interval of the model's forecasts.

Having established the model's efficacy for predicting future EV counts, we proceeded to utilize the optimized parameters to construct a final model encompassing the entire observed dataset (without splitting it into training and testing sets). This final model serves as the primary tool for forecasting future EV adoption trends in the different counties.

## IV. RESULT AND DISCUSSION

### A. Electric Vehicle Adoption Trends

Our analysis of data from the Washington State Department of Licensing (DOL) API revealed a significant upward trend in electric vehicle (EV) registrations over the past decade. This consistent exponential growth signifies a substantial and growing demand for EVs within the state. This trend presents a compelling opportunity for companies involved in EV charging infrastructure development and investment.

### B. Spatial Distribution of EVs and Charging Stations

While raw EV counts provide valuable insights, a more comprehensive understanding requires examining the spatial distribution of EVs and charging stations. Notably, King County boasts roughly half (approximately 50%) of the 1,686 charging stations across Washington. This concentration contrasts with counties like Island, Clark, Skagit, and Whatcom, which have significantly fewer stations. Based solely on this ratio, these latter counties might appear attractive for targeted investments in charging infrastructure. However, it's crucial to consider additional factors beyond raw numbers.

### C. Model Performance and Limitations

The time series models employed in this study, particularly the Seasonal ARIMA (SARIMA) model, demonstrated promising results in forecasting future EV adoption trends within Washington counties. However, it's important to acknowledge the inherent limitations of such models. They rely on historical data and may not perfectly capture unforeseen events or sudden shifts in consumer behaviour. Additionally, data quality plays a crucial role in model accuracy, and any inconsistencies or missing values can potentially affect the forecasting capabilities.

Below are the result graphs depicting the forecasted trends in electric vehicle (EV) charging demand across Washington counties, providing valuable insights for strategic infrastructure planning and future charging station placement.

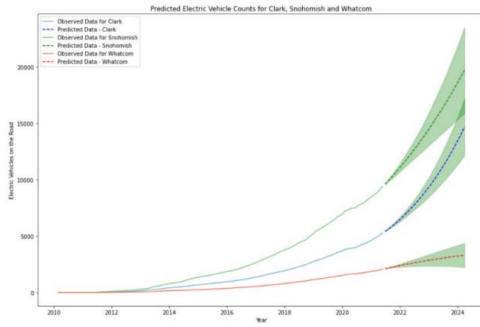


Fig. 2. Analysis on Clark, Snohomish, and Whatcom Counties

**Clark, Snohomish, and Whatcom Counties:** These three counties show the highest potential for the EV charging business. So, we advise electric vehicle charging companies to make investments in the counties of Clark, Snohomish, and Whatcom.

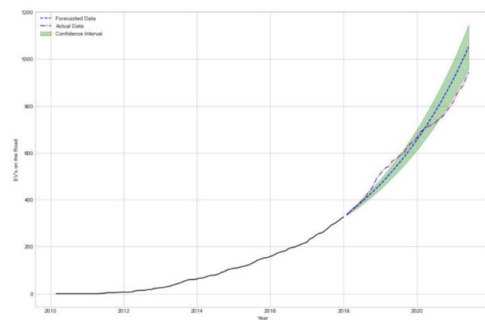


Fig. 3. Analysis with Island County

**Island County:** As we can see above, the model is predicting that the electric vehicle counts in Island County will keep growing exponentially in the coming years.

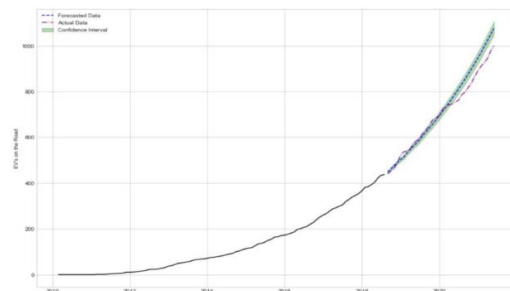


Fig. 4. Analysis with Benton County

**Benton County:** The graphs above depict the forecasted trends in electric vehicle (EV) adoption for various counties across Washington State.

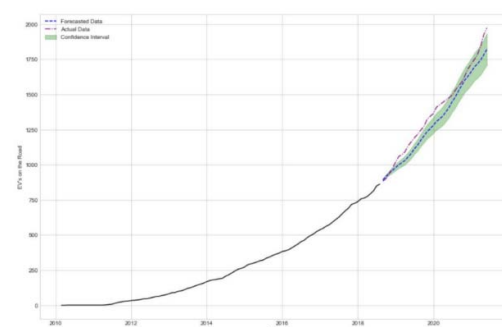


Fig. 5. Analysis with Spokane County

**Spokane County:** Above, we can see that the model is predicting the electric vehicle count in Spokane County to keep increasing exponentially.

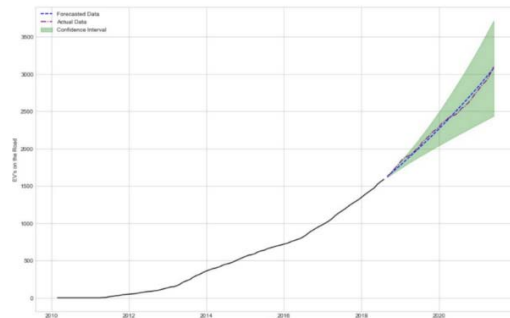


Fig. 6. Analysis with Kitsap County

**Kitsap County:** Above, we can see that the model is predicting the amount of electric vehicles in Kitsap County to keep increasing exponentially.

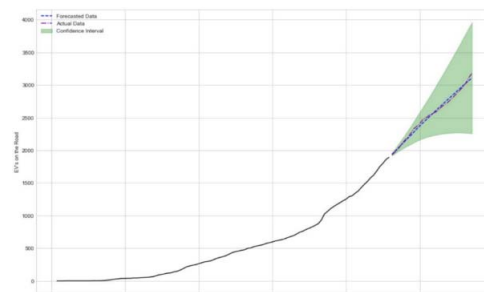


Fig. 7. Analysis with Thurston County

**Thurston County:** The model is predicting that the amount of electric vehicles in Thurston County are going to keep increasing at a linear rate (based on the mean prediction).

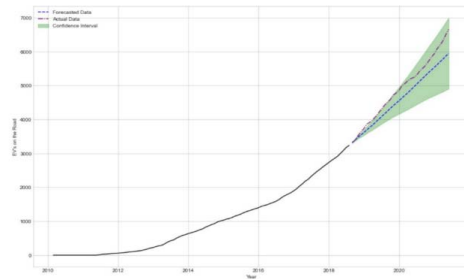


Fig. 8. Analysis with Pierce County

**Pierce County:** This model is predicting that the growth in electric vehicles in Pierce County will keep an exponential growth pattern for the next few years.

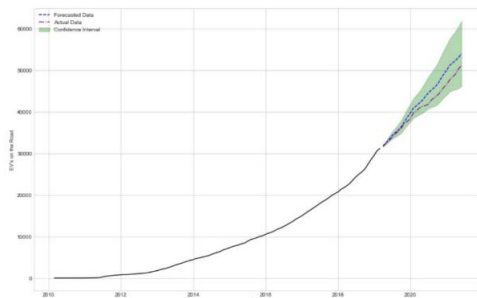


Fig. 9. Analysis with King County

**King County:** The model is predicting that the amount of electric vehicles in King County are going to keep increasing almost linearly based on the mean values. If we look at the upper confidence interval though, we can see that the increase could keep an exponential pattern.

## V. CONCLUSION AND FUTURE WORK

With the increasing utilization of Electric Vehicles (EVs), driven by technological advancements and supportive policies, investing substantially in charging infrastructure has become imperative. This research study indicates a consistent, exponential rise in EV adoption over the past decade, a trend anticipated to persist. While our study has centred on the top 10 counties within Washington, future investigations could extend to encompass all counties and even delve into city and zipcode-level analysis for enabling refined charger placement strategies. Potential avenues for expansion include integrating geospatial data for enhanced model accuracy, exploring alternative charging technologies, assessing policy impacts on EV adoption and infrastructure, analysing the used EV market, studying commuting patterns, and expanding the analysis to other regions for broader insights. By pursuing these avenues, stakeholders can gain deeper insights into EV adoption trends and charging infrastructure requirements, facilitating informed decision-making and sustainable transportation electrification.

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